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BIOMEDICAL LIMITED

SITE MASTER FILE (SMF)

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BIOMEDICAL LIMTED
KWARA STATE, NIGERIA

CONTROL COPY

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Approval and Authorization

	WRITTEN BY	REVIEWED BY
Signature:		
Position:		
Date:		

	APPROVED BY	AUTHORIZED BY
Signature:		
Position:		
Date:		

1.0 GENERAL INFORMATION

1.1 Brief information on the firm

BIOMEDICAL LIMITED is an indigenous Pharmaceutical Manufacturing Company specialized in the production of intravenous solutions in flexible PP containers under Bioflex brand, and in the production of Oral Liquid preparations (syrups and suspensions).

Non-pharmaceutical formulations manufactured under the banner include hand sanitisers, handwash and dishwash preparation. The plant is operated as a low-risk environment, specifically handling only pharmaceutical materials, excluding materials that pose severe, immediate or chronic health risk to workers and the environment. Veterinary products are not manufactured in this site. Incorporated in 1978, the company commenced commercial production on 21st August 1981. The company sustains the quality of its products by a joint effort between participating departments, spearheaded by the actions of the Quality Assurance department.

1.2 Pharmaceutical manufacturing activities as licensed by the national authorities.

1.2.1 The company was registered by Corporate Affairs Commission on 13th day of April 1978.

1.2.2 Licensed year by Pharmacist Council of Nigeria; with the current version being 2026.

1.2.3 With GMP observance certification, products are registered with NAFDAC and renewed every 2 years.

Copies of licensing shown in Annexure 1

1.3 Name and Address of registered office and factory

Name: Biomedical Limited.

Address: No. 1, Ohimege Road, Gaa-Imam Industrial Estate, PMB 1449, Ilorin, Kwara State, Nigeria.

Phone No: 08138853131, 09027475550, 08079409494, 08034814697.

E-mail: info@biomedicalng.com

1.4 Contact persons:

Name	Mr. Muthukumarraja Chidambaram	Mrs. Toyin Utomakili
Position	Managing Director	Executive Director
Email	muthukumararajac@gmail.com	toyinuto@yahoo.com
Mobile No	+22893015174	+234 805 700 1168
Name	Mr. Abdurauf Tiamiyu	Mr. Agboola Liadi
Position	Executive Director	Factory Manager
Email	ratiyu@yahoo.co.uk	agboolaadetunji@yahoo.com
Mobile No	+234 806 077 5603	+2347064351983

1.5 Products Manufactured at the Site

Large-Volume-Parenteral manufactured at the site include:

S/N	PRODUCTS	NAFDAC No.	PACK SIZE
1	Sodium Chloride 0.9% $^{w}/_{v}$ (N, NL)	04-4893	500ml, 1000ml
2	Dextrose 5% $^{w}/_{v}$ (D, DL)	04-4894	500ml, 1000ml
3	Dextrose 5% $^{w}/_{v}$ + 0.9% $^{w}/_{v}$ Sodium chloride (GS, GL)	04-1342	500ml, 1000ml
4	Dextrose 10% $^{w}/_{v}$ (10D)	04-4895	500ml
5	Dextrose 4.3% $^{w}/_{v}$ + 0.18% $^{w}/_{v}$ Sodium Chloride (4G)	04-4896	500ml
6	Isoplasma 4% (IS)	04-1334	500ml
7	Biogyl® Metronidazole injection (B)	04-1343	100ml
8	Darrow's Solution Full Strength (DF)	04-1322	500ml
9	Darrow's Solution Half Strength (DH)	04-4897	500ml
10	Ringer Lactate Solution (R)	04-4892	500ml
11	Mannitol 10% (1M)	04-1333	500ml
12	Mannitol 20% (2M)	A11-0134	500ml
13	Dextrose 50% (50G)	A11-0443	100ml
14.	Ciprofloxacin injection (CP)	A11-1190	100ml

15.	Fluconazole injection (FL)	A11-1150	100ml
16.	1.5% CAPD (PD)	A11-1152	2000ml
17.	2.5% CAPD (2PD)	A11-1151	2000ml

In addition to the large volume intravenous fluids currently manufactured in the premises, the company has an oral liquid plant that manufacture essential generics products with clearance obtained from National Agency for Food and Drugs Administration and Control Agency. The plant began operations on the 20th September 2020. The following are products currently manufactured in the Oral Liquid plant.

S. No.	Products	NAFDAC No.	Pack size
1	Biomedical Paracetamol Syrup (BL)	A11 - 1164	60 mL
2	Biomedical Cough Expectorant (BC)	A11 - 1166	100mL
3	Biogyl Suspension (BM)	A11 – 1165	60mL
4	Ulcamed Suspension (US, 2US)	A11 – 1168	100mL/200mL
5	Biobabe Syrup (BB)	A11 – 1171	60mL
6	Basafen Suspension (BE)	A11 – 1167	100mL
7	Biomedical Vitamin C Syrup (CS)	A11 –1170	100mL
8.	Chloroquine Syrup (FS)	A11 – 1169	60mL
9.	Bioasth Syrup (BA)	A11 – 100058	100mL
10.	Biothazine Syrup (PZ)	A11 – 100049	60mL
11	Basaton Syrup (BN)	A11 – 100057	100mL
12.	Paraton Syrup (PN)	A11 – 100056	100mL
13.	Bradvite-M Multivitamin Syrup (BD)	A1 - 100048	100mL
14.	Ex-Coff Syrup (EX)	A11 – 100047	100mL
15.	Albmed Suspension (AB)	A11 - 100112	10mL
16.	Flexron Blood Tonic (FN)	A11 – 100105	200mL
17.	Ipec H Syrup (IH)	A11 - 100045	100mL

18.	Biomedical co-trimoxazole suspension (BX)	A11 - 100108	50mL
19.	Coflex-B Multivitamin Syrup (CB)	A11 - 100107	100mL
20.	Magflex Suspension (MG)	A11 - 100106	100ml/200mL

1.6 Short description of the site (size and location)

Biomedical Limited is located at the industrial estate, Gaa-Iman area of Ilorin, Kwara state, North central region of Nigeria. The total area of the plot is 2.5 Hectares, while the total built up area is 3650Square meter.

Immediate environment of the site is as follows:

- To the East: Residential area
- To the West: Rajrab Pharmaceutical Limited
- To the South: industrial area
- To the North: Winners Chapel Int'l church
- Company Coordinates: 8.4629761, 4.5630350.

1.7 Workforce

The company has a total manpower of 406 made up of 100 permanent staff. Making up the 406 personnel are ad-hoc workers distributed across various departments in the company. The breakdown are as given below.

Department	Management	Senior Staff		Workmen		TOTAL
		Infusion	Oral Liquid	Infusion	Oral Liquid	
Plant Management	Executive Chairman					1
	Managing Director					1
	Executive Director					1
	Assistant General Manager					1
	Factory Manager/ Superintendent Pharmacist					1
	Production Manager					1
	Quality Assurance Manager					1
	Chief Engineer					1
	Admin Manager					1
	Audit Manager					1
	National Sales Manager					1
	Marketing Manager					1
	Quality Control Head					1
	Documentation Head					1
Production		10	4	156	88	258
QA/QC		8	5	2	2	17
Stores		2	1	5	4	12
Transport & Logistics		1	-	4	2	7
Engineering		10	5	6	3	24
HR/ Admin		8	2	20	8	38
Finance/ Accounts/ Audit		3	2	1	1	7
Sales and Marketing/Credit control		20	5	3	1	29
TOTAL		406				

1.8 Quality Management System (QMS)

1.8.1 Quality Policy and Vision

In Biomedical Limited, we have a strong commitment to continual production of quality pharmaceuticals through global best practice. Our quality system is designed to satisfy our customers and meet all regulatory expectations.

This quality policy is driven by our company commitment to, employing highly skill and well-motivated staff, continual training, development and improvements in order to achieve our set objectives.

Full definition of the company's quality policy is given in the Quality Manual (BML-QA-QM-001 – Annexure 2)

1.8.2 Short description of the Quality Management System.

Quality Management System definitions ensure compliance with cGMP requirements as enforced by regulatory imperatives. The implementation of QMS is defined in the Quality Manual, which outlines "Quality Policy" and defines requirements of systems necessary to assure product quality. The intent of the Quality Manual is to state what system needs to be in place and who is responsible for implementation of these systems. The purpose of this Quality Manual is to define the Quality Management Principles and the quality system elements of BIOMEDICAL LTD ensuring that the end product meets all predefined specifications regarding identity, strength, safety and purity of the products. The organizational structure is such that technical, administrative and human factors affecting quality are controlled and reviewed by management.

The Quality Management System functions in such a manner so as to provide proper confidence that

(a) The system is well understood and effective as per cGMP

(b) The products or services satisfy the customer's expectations fully.

(c) Continually improving the effectiveness of QMS through motivated and trained workforce.

(d) Emphasis is laid on problem prevention rather than dependence on detection after occurrence

The quality system typically applies to and interacts with all activities pertinent to the quality of a product or service. It involves all phases from initial identification to final satisfaction of requirements and customer expectations starting from marketing and market research, design/specification engineering and product development, procurement, process planning and development, production, testing and examination, packing and storage, sales and distribution, installation and operation of the machines technical assistance and maintenance. Stability studies are performed as per regulatory and marketing imperatives.

For the organization's mission, vision and quality policy, see annexure 2.

The company has a documented quality system to ensure a high degree of product quality. The salient components of the Quality System includes;

- cGMP / cGLP enforcement, implementation and compliance.
- Preparation, approval and control of Standard Operating Procedures (SOPs)
- Preparation, approval and control of Specifications
- Preparation, approval and control of Standard Test Procedures (STPs).
- Approval of Master Batch Formula records
- Control of Quality Documents and release of batches e.g. Batch Production and Control Records
- Documented sampling procedures for raw and packaging materials, in-process materials and finished products
- In-process quality controls
- Calibration of instruments
- Preventive Maintenance of equipment and instruments
- Preparation, approval and coordination of Validation Master Plan comprising of
  - Facility Validation: HVAC validation, water systems, compressed air, etc.
  - Equipment qualification: Design, Installation, Operational and Performance Qualification
- Process Validation comprising of
  - Prospective
  - Concurrent
  - Retrospective

- Analytical Method Validation
- Environment monitoring
- Cleaning Validation and disinfectant validation
- Conducting and Coordinating training activities
  - Induction
  - Current Good Manufacturing Practices
  - Standard Operating Procedures
  - On-Job training
- Vendor Development and Approval : Vendors (manufacturer and supplier, where appropriate) for the raw and packaging materials are approved as per approved SOP which describes the procedure to evaluate the concerned vendor(s),on pre-purchase samples and manufacturer's and/or supplier's site visit as per requirement.
- Audit Programs
  - Vendor for raw and packaging materials
  - External calibration laboratory
  - Contract Analysis
  - Internal Audits and Self Inspection: Department wise internal audits and self-inspection are conducted by the competent staff, as per the written approved SOP. The CAPAs are recorded against the discrepancies observed/ found and audit report is submitted to the top management.
- Review of Batch Production and Control Record and Release of Finished Goods are undertaken as per the approved procedures.
- Handling of complaints from customers or regulatory agencies
- Control (reserve) samples of raw materials and finished goods
- Handling of product recalls
- Conducting annual product review.

A flow diagram of major aspects of quality system is attached as an **Annexure 3**.

2.0 PERSONNEL

2.1 Organogram

Organizational chart showing the arrangement for every department is enclosed herewith. See **Annexure 4**.

2.2 Qualification, Experience and Responsibilities of Key Personnel

For qualification & experience of the key personnel refer **Annexure 5.**

2.3 Elements of the Quality Assurance System.

The Quality Assurance Head is mainly responsible for incorporating comprehensively designed and correct systems of cGMP, aimed at producing safe, pure and effective products. However, owing to the shared responsibility of QMS, the following are the responsibilities of the key personnel.

Job responsibilities: Head – Production.

- Head of unit.
- To ensure the development, implementation and adherence to current SOPs. Training of subordinates in SOPs.
- Supervision, and the allocation of processing resources, ensuing compliance to cGMP and safety requirements.
- Provide assistance in documentation related to production and cGMP.
- Maintenance of good housekeeping, cleanliness, personnel hygiene, and work discipline.
- Provide support and coordination regarding activities like inspections, validation, qualifications, planning, and production engagement activities.

Job Responsibilities: Quality Assurance – Head

- Documentation for all QA activities.
- Daily monitoring of all production and related activities.
- Training on cGMP and personnel hygiene.
- Reviewing the Master records, Batch records, SOPs and protocols.
- Approval of vendors.
- Engages the periodic audits of the quality systems.
- Stability studies of finished products.

- Coordination with various departments.
- Actively provides direction during audits.
- Product releasee and recall.
- Validation and qualification
- cGMP trainings.

Job Responsibilities: Engineering – Head.

- Observance of Good Engineering Practices in the plant.
- Aiding and ensuring compliance with cGMP.
- Design, implement and maintain the cleanroom concepts.
- Coordinate with the Production and Quality Assurance regarding productivity.
- Spearhead the energy conservation and cost minimization mechanisms.
- Development of systems and methodologies in line with regulations.
- Assist during audits.

2.4 Outline of arrangements for basic and in-service training and how records are maintained:

For the systemic development of the knowledge, attitude and skill behavior required by an employee, the company has identified training as an important tool. The need for training is identified by the department head after observing and reviewing the performance during appraisal of the employee. As per need, in-house and external training is given to the employee. The procedures and guidelines for execution of these trainings are specified in the SOP and Training Matrix (BML-QDG-QA-108-L01). Four types of training are provided to the personnel - Induction Training, cGMP / cGLP Training, SOP Training and On-Job Training.

2.4.1 Induction Training:

All new employees are given Induction Training to familiarize the new employee with the organization he or she is joining. Induction Training Manual () is used for giving the training. The training covers topics like a brief history of the company, organization of company, a brief product profile and administration & human resource function etc.

2.4.2 *cGMP / cGLP Training:*

All employees, staff and workers are given cGMP / cGLP training relevant to his or her work. The cGMP / cGLP training are based on regulatory guidelines such as WHO, EU GMP guidelines and Ph Eur etc. cGMP / cGLP training includes topics important for maintaining product quality and integrity, contemporary topics, revisions to regulatory subjects, etc.

2.4.3 Training on Standard Operating Procedures:

All staff members are given training on various Standard Operating Procedures relevant to their area of work.

2.4.4 On-Job Training:

On-job training is conducted by respective department heads on various topics relevant to their area of work.

2.4.5 Retraining:

Retraining needs are identified by assessments of questionnaires or as a result of audit deficiencies noticed. Every employee is retrained on routine activities after certain period.

As per the requirement a questionnaire is given to the trainee, which is evaluated by the trainer to identify the efficacy of the training. The records of the training are stored and maintained by Quality Assurance documentation unit.

2.5 Health requirements for personnel engaged in production.

Employees undergo medical checkup twice a year at an approved Hospital/Diagnostic health facility. Health and medical records of each personnel engaged in Production and analysis are maintained. There is a documented procedure for Medical Examination. Three main types of medical examinations - Medical Examination at the time of joining, Regular Medical Examination, Medical Examination after re-joining after a medical leave, are conducted to ensure that each person is in good health. Any personnel with any communicable or contagious disease or open lesions or wounds are not allowed to join duty.

All health records are maintained by Human Resource department.

2.6 Personnel Hygiene requirements, including clothing:

Personnel are trained to maintain a high level of hygiene. A common change room is provided for all personnel, equipped with adequate shower facilities, to ensure proper

personal hygiene before entry into controlled production areas. A dedicated change room is provided for visitors. Lockers are provided in the change rooms for keeping personnel belongings. All workers and staff are required to change the street cloth, take their shower and then wear clean dress as per requirements and standard procedures for entry and exit. Smoking, eating, drinking, chewing, etc. are not permitted in production, laboratories, stores and other areas, which might adversely affect the product quality. Area-wise colour coding of uniforms for employees and visitors is implemented and maintained to ensure proper segregation and prevent cross-contamination. Washing of uniforms is carried out using an in-house decontamination and laundry system in accordance with approved procedures.

3.0 PREMISES, FACILITIES AND EQUIPMENT:

3.1 Simple plan or description of manufacturing areas drawn to scale

All the plant layouts are enclosed, Refer the Attached **Annexure 6**.

3.2 Nature of construction and finishes

The building is of reinforced cement concrete construction. Internal and external walls are finished with smooth, impervious, non-shedding, and paint suitable for pharmaceutical use. Wall panels are installed to segregate clean areas as per functional and cleanliness requirements. Each segregated area is supported by a dedicated HVAC system to maintain the required environmental conditions and prevent cross-contamination. Corners in clean areas are designed with proper covings. Flooring in all manufacturing areas is epoxy-coated, seamless, and non-shedding. In critical manufacturing and primary packaging areas, all fittings and fixtures including doors, windows, light fittings, AHU air inlets, and exhaust outlets are flush-mounted and integrated symmetrically with walls and ceilings. This design ensures smooth, continuous surfaces without ledges or crevices, thereby preventing dust or powder accumulation and facilitating effective cleaning and sanitation.

3.3 Water Treatment System

3.3.1 Brief description of water treatment.

Raw water sourced from a borehole is treated to produce Purified Water and Water for Injection in compliance with the requirements of the European Pharmacopoeia. Raw water is collected and stored in a 10,000 L overhead storage

tank and transferred to the water treatment plant through high-density PVC piping. A leachability and extractables study was conducted on the piping material and found to be compliant, with no detectable toxic substances. The corresponding report is provided in Annexure 7, reference SE/0197.

The raw water first passes through a sand filter designed to remove suspended solids, debris, and coarse particulates. The filtered water then flows through an activated carbon filter, which further clarifies the water by decolorization and deodorization, and removes residual chlorine, organic contaminants, and pesticide residues. The treated water subsequently passes through a water softener unit, where hardness is reduced by ion exchange. Calcium and magnesium ions are replaced with sodium ions, thereby preventing scale formation and calcareous precipitation within downstream equipment and piping. When the ion exchange resin becomes exhausted, regeneration is performed using a saturated sodium chloride solution. During regeneration, calcium and magnesium ions are displaced from the resin and discharged to drain, restoring the resin's sodium exchange capacity.

The softened water is then passed through 0.5 micron cartridge filters prior to entry into the demineralization system, followed by reverse osmosis. The reverse osmosis system removes dissolved ionic species, organic compounds, microorganisms, and suspended impurities through semi-permeable membrane filtration. Under applied pressure, purified water permeates the membrane while dissolved and suspended contaminants are rejected and concentrated in the reject stream.

Chemical sanitization of the reverse osmosis membranes is performed using hydrogen peroxide according to validated procedures. The associated piping between the reverse osmosis unit and the storage tanks is also chemically cleaned as part of the sanitation process. Reverse osmosis water is subsequently fed into a multi-column distillation unit to produce sterile, pyrogen-free Water for Injection. The distilled WFI is collected and stored in a dedicated WFI storage tank. The WFI distribution system is engineered to maintain a minimum circulation velocity of 1 m/s during continuous recirculation, even under no-use conditions,

to prevent microbial proliferation and biofilm formation. The WFI storage tank is designed for atmospheric operation and is fitted with a hydrophobic vent filter to prevent tank pressurization while maintaining microbiological integrity.

During normal operation, the conductivity of the WFI is continuously monitored using an inline conductivity cell installed in the return loop of the distribution system to ensure ongoing compliance with pharmacopeial specifications. The total storage capacity of the WFI system is 8,000 L. Water stored in the WFI tanks is maintained under continuous recirculation to preserve chemical and microbiological quality.

In the Oral Liquid manufacturing block, a dedicated water treatment system is installed to produce Purified Water BP used in oral liquid manufacturing. The treatment train mirrors the design and operational principles of the Infusion Water Treatment Plant up to the stage of demineralisation, terminating at the generation of Demineralised Water that complies with the specifications of Purified Water BP.

The system comprises pre-treatment units including sand filtration, activated carbon filtration, water softening, and cartridge filtration, followed by an ion exchange process to achieve the required chemical purity. Unlike the infusion plant, the process does not proceed to further purification steps such as distillation or WFI generation. The final treated water consistently meets pharmacopeial requirements for Purified Water, including defined limits for conductivity, total organic carbon, microbial count, and absence of specified contaminants. Monitoring systems are installed for critical parameters such as conductivity and flow rate, with routine chemical and microbiological testing performed according to approved SOPs.

The purified water is stored in a dedicated sanitary storage tank and distributed through a closed-loop system constructed of appropriate pharmaceutical-grade materials. The loop is designed to minimize stagnation and microbial proliferation. Where applicable, periodic sanitization is carried out using validated procedures to maintain system control. This Purified Water BP serves as the starting material for oral liquid formulations and is released for use only after

confirmation of compliance with established in-house and pharmacopeial specifications.

Refer **Annexure 7** for schematic drawing of water system.

3.3.2 Material of Construction of Vessels and Pipe Work:

WFI water is supplied to production areas through SS 316L pipelines. Storage tank is mechanically polished and jacketed outside and made up of SS 316L. The storage tank is equipped with level transmitter and vent filter. See Annexure 7 for MOC report

3.3.3 Validation Stage:

Functional design specifications are checked and evaluated against user requirement. Subsequently IQ/OQ of the water system is performed as per the regulatory requirement.

This PQ Study must be executed in three phases:

Phase 1

The phase-1 will last for minimum two weeks for testing of water of every stage to meet the specifications, during which preliminary sampling of water shall be done to assure that the system is ready to go into Phase- 2, the prospective phase of validation.

During this phase operating, regeneration and sanitization procedures shall be verified & frequencies for sanitization shall be established and fixed.

On completion of Phase-1, report shall be prepared and it shall be checked and approved for further proceeding to Phase-2.

Phase 2

Phase-2 carried out for Two weeks with sampling frequency like Phase-1.

Phase 3

Phase 3 shall last for about Twelve (12) months with each sampling point being covered every week.

Post Validation Stage:

Once validation is completed, the water system shall be maintained in a state of control by keeping the system within the acceptable ranges for the key control parameters established during the validation program and by adhering to the

validated operating and maintenance procedures. These key control parameters shall be: flow rate, temperature, conductivity, pH, etc.

Any change to the validated water system may or may not invite validation (depending upon the effect of the change on the output of water system). The change shall be processed through change control procedure.

3.3.4 Water For Injection

Water for injection is produced, stored and distributed in a manner which prevents microbial growth by constant circulation at temperatures not less than 70 °C. The qualification of the Multicolumn Distillation Plants (MCDP) be performed and evaluated against the requirement of use.

The validation of water for injection system shall be performed through an approved protocol. Validation shall preferably be completed in three phases; to be followed over a period of a year, as described for purified water system validation.

3.3.5 Maintenance (Description of Planned Preventive Maintenance Programmes and recording system).

A well-planned preventative maintenance procedure is in place to avoid any breakdowns. Detailed procedures for carrying out maintenance activities and services being carried out by the engineers are defined in respective SOPs for water system, ventilation system etc. describing the preventive maintenance operations. A maintenance program is strictly followed strictly as per approved planner to ensure designed output from machinery. Records of any repair and maintenance of building are maintained by Engineering Department. Any adverse findings are reported to the respective users by Engineering-In-Charge.

3.4 HVAC Systems

3.4.1 Brief Description of the HVAC System

The HVAC system consists of a recirculation-type Air Handling Unit (AHU), floor-mounted and constructed from double-skin panels insulated with 50 mm polyurethane foam of density 40 ± 2 kg/m³. The AHU comprises distinct functional sections including a mixing chamber, blower section, DX cooling coil section, and multi-stage filtration section. The filtration system includes a fresh air pre-filter

(10 micron), a pre-filter (10 micron), a fine filter (3 micron), and a bleed air filter (3 micron). Conditioned air is distributed to the respective rooms through a ducted air distribution system and supplied into the rooms via terminal HEPA filters to achieve the required air cleanliness levels. Manually operated AHU dampers of low-leakage aerofoil design are provided to regulate airflow. A fusible link-type fire control damper is installed to automatically shut off air distribution in the event of fire. Magnehelic pressure gauges are installed across filters to monitor differential pressure and detect filter clogging. The AHU is designed to operate on a power supply of 415 ± 10 VAC.

The Cleanrooms are classified based on airborne particle concentration in accordance with ISO 14644-1 guidelines. To prevent cross-contamination and to maintain specified temperature and relative humidity conditions in manufacturing areas, the HVAC system is utilized.

There are four rooms currently subjected to treated air

- Dispensing room
- Filling room
- Batch coding room
- Clean room corridor
- Microbiological room

The temperature and humidity sensors provided in the HVAC system are calibrated periodically as mentioned in the respective written procedure (SOP – BML/ENG/NG/003).

Temperature and humidity are controlled and maintained within the limits as per the requirement. The pre-filters are cleaned as per the frequency described in the SOP for cleaning of the filters.

In case of pressure differential failure, all the production activities are stopped, and the filters are replaced with cleaned filters.

Validation of the HVAC system (as defined in BML/QA/HVAC/PT/001)is done through the studies of parameters like air velocity, air change per hour, temperature, relative humidity, non-viable particle count, HEPA filter integrity, pressure differential across HEPA filter, Air flow visualization, recovery test,

viable particle count. In processing area daily monitoring of differential pressure, temperature and relative humidity, non-viable particle count is carried out. Microbial monitoring is carried out in area as per requirement (SOP - BML/QDG/MC/010 and BML/QDG/MC/011).

Each of these rooms has the following features

- **HEPA** (High Efficiency Particulate Air) Filters- this is part of the HVAC system and is a tertiary filter. It is a series of matted filters of a 0.3micron that excludes all contaminant (particles and microbes) that size.
- **Air Exhaust**- these are the outlet for the ambient air in the clean room
- **Dynamic Pass Box**- (Pouch Transfer door) – this is a device used to aseptically transfer material between clean rooms
- **Differential pressure gauges**- differential pressures between any two adjacent areas shall be at least 15pascal (0.06 in inches (or) 1.5mm water gauge). Suitable pressure differential gauges are used to measure pressure differential between rooms.
- **Air Shower**- this is a specially modified doorway that uses clean, treated air to hygienically clean the garments of individuals before they enter a clean area. The air shower should rid clothing and skin of any surface particles and reduce microbial load. This is the main entrance into the filling room
- **Air Lock**- this is a system that uses a pressure differential between two rooms with different levels of cleanliness. The pressure differential prevents the mixing of air between rooms and is maintained using doors that keep the air "locked" in place.

The Oral liquid plant uses the AHU to achieve the corresponding environment classified in ISO 14644-1 guidelines. The AHU supplies pre-treated air into the key rooms in the production areas through ductless low diffusion inductor to prevent contamination and maintain the relative humidity and temperature warranted for the environment.

3.5 Production and Quality Control Equipment:

3.5.1 Brief description of Production equipment:

All the production equipment are designed to meet GMP requirements. All contact parts of the equipment are constructed of Stainless Steel 316 grade. Equipment are qualified for their Design, Installation, Operational and Performance. All equipment are placed to suit the operations to be carried out. The layout and design of equipment is aimed at minimizing the risk of errors and permit cleaning and maintenance to avoid cross contamination or any adverse effect on the quality of products produced. All major equipment are facilitated with CIP (Clean in Place) System. Analytical Balances and other measuring equipment are of an appropriate range to achieve accuracy and precision. Some of the major equipment are as follows: Automatic filling machines, Dispensing and Mixing-vat systems, Moist heat sterilisers etc.

3.5.2 Brief description of Quality Control equipment:

The Quality Control Department is organized into two dedicated laboratories, namely the Infusion Laboratory and the Oral Liquid Laboratory. Each laboratory primarily supports testing and quality control activities for its respective manufacturing plant. Both laboratories are equipped with appropriate and sophisticated analytical instruments, including but not limited to UV-Visible Spectrophotometers, High Performance Liquid Chromatography systems, analytical balances, pH meters, conductivity meters, incubators, laminar air flow benches, melting point apparatus, flame photometers, stability chambers, viscometers, as well as auxiliary laboratory equipment required for routine and specialized analyses.

All critical and sophisticated analytical instruments are qualified and validated in accordance with approved procedures and are subject to routine calibration and preventive maintenance programs. Calibration and validation activities are performed at defined intervals, with documented records maintained to ensure data integrity, accuracy, precision, and ongoing compliance with GMP and applicable pharmacopeial requirements. The microbiology laboratory is facilitated with separate environment for determining the microbiological purity of water, raw material, environmental assessment and finished products.

Computers are used for documentation purposes and data entry at different departments. Access to department computers is restricted by using suitable passwords. A consolidated list of equipment is attached as **Annexure- 8**.

3.5.3 Maintenance (Description of planned preventive maintenance programme and recording system)

Regular preventive maintenance is undertaken in all the areas for various equipment/facilities as per written procedures (SOP – BML/ENG/NG/003) and records are maintained thereof. All details including maintenance schedules, records of services or breakdown and any modifications are recorded.

Engineering department is responsible for maintenance and servicing of the machinery. Engineering personnel have a scheduled preventive maintenance program for all production machinery and utilities. Maintenance routine check-ups, which could affect the product quality, are clearly identified. If any major changes are made regarding equipment, then the equipment is revalidated.

3.5.4 Qualification and Calibration, including the recording systems and arrangements for computerized system validation:

All equipment are qualified (DQ, IQ, OQ and PQ) before use. Records of calibration/validation (SOP – BML/QDG/QA/031 and BML/QDG/QA/111), which are maintained in prescribed formats, are reviewed periodically. A Validation Master Plan (VMP – BML/QA/VMP/002) is designed for validation of facility, utilities, process and cleaning operations. The validation plan is designed to demonstrate that the quality features are built into the facility, utilities and processes ensuring that they are fully functional, remain in place and conform to the relevant regulatory requirements. Equipment re-validation is carried out if there is any change(s) or after major breakdown in the major component that could have a considerable impact on the performance of the equipment. Additionally, all the critical equipment are subjected to routine revalidation.

All validation activities are performed in accordance with the approved protocols.

Equipment calibration:

All the equipment and instruments are calibrated as per the calibration planner before use as per the procedures and frequency mentioned in the respective

SOPs. And then the calibration tags mentioning the calibration status and details of equipment are affixed to the equipment. All the equipment are calibrated by both in-house and by pre-approved external agencies with standards traceable to some national or international agencies as per SOP BML-QDG-QA-031.

Computer and software validation:

Control systems and PLCs associated with manufacturing equipment are validated in accordance with the approved protocols as per the guidelines mentioned in VMP.

3.6 written specifications and procedures for cleaning and sanitization of manufacturing areas and equipment.

The manufacturing premises is cleaned and maintained in an orderly manner and there is chance to accumulated waste, dust and debris are disposed in timely and sanitary manner. SOPs are available describing responsibility, cleaning procedures and sanitization procedure/ methods / specifications, cleaning agents used and schedule. Rodenticide and insecticides are used to keep rodent, insect and other pests under control.

Written SOPs are available for cleaning of equipment, vessels, etc. Records of cleaning are maintained. Adequate measures are taken to ensure that clean equipment; vessels are not contaminated prior to use.

Disinfectants are rotated / changed as per the schedule mentioned in the SOP. Bioburden is monitoring and equipment cleaning validation studies are used for evaluation of effectiveness of cleaning procedures. equipment are subjected to swab / rinse tests for chemical analysis and areas are subjected to swab test / active and passive air sampling / plate exposure methods for microbial counts.

Cleaning of water supply systems including storage tanks is done as per written procedures. Pre-filters and fine filters of air handling system are cleaned at defined frequency or schedule.

The pest control activities are done and recorded as per written procedures mentioned in the SOP and only substances approved for use in Pharmaceutical industry are used. Insect / rodent traps are also placed at suitable points to control the insect / rodent infestation in the area.

4.0 DOCUMENTATION

4.1 Arrangements for the preparation, revision and distribution of necessary documentation for the manufacture.

Well-established documentation system exists to implement Good Documentation Practices, which is a part of Quality Assurance. Clearly written documents are prepared, reviewed and controlled, which prevents the errors from verbal communication and permits traceability for all activities carried out as we have it in Document and data control (SOP- BML-QDG-QA-002). Quality Assurance documentation unit is responsible for the revision and distribution of documents and concerned departments are responsible for preparation and initiation of changes in procedure.

Master documents are stored in the archives of Quality Assurance department and controlled copies are distributed to the concerned departments.

The documents are designed, prepared, reviewed and controlled as per the SOP(s).

Documents are approved and authorized, signed and dated by the designated persons. The documents specify the title and nature, also mention the purpose, if required. They are laid in an orderly fashion. Reproduced documents are clear and legible and are appropriately stamped, whether electronically or physically. These are normally reviewed as per the review period mentioned in the relevant SOPs.

An SOP is available for correction of a wrong entry in a record (SOP – BML/QDG/QA/101). The batch production and control records are completed at the time of each operation in such a way that all significant activities concerning the manufacture of the product are traceable.

Quality documents and batch records are retained as per SOP (BML/QDG/QA/004), from the date of obsolescence or date of expiry of the relevant drug product. SOP is available for issuance, traceability during storage, retrieval and destruction of records.

Any change in the approved documents/equipment is processed through a system of Change Control (SOP – BML/QDG/QA/071).

Different types of master documents and records available with company are listed below:

- Site Master File

- Quality Manual
- Master Batch Formula
- Batch Manufacturing Records
- Validation Master Plan
- Validation Protocols and Reports
- Stability Protocol and Reports
- Standard Operation Procedures
- Specifications (Raw Material, In-process Material and Finished Product)
- Standard Test Procedures

All the documents are controlled by Quality Assurance Documentation unit of Quality Department. The system is designed to ensure that all the documents are properly distributed, controlled and stored so as to maintain their integrity. The access to stored documents is limited to the authorized persons only and all documents are stored in such as way so that they are easily retrievable.

4.2 Any other Documentation Related to Product Quality which is not mentioned elsewhere.

Other related documents are also available or to be available. These are:

- Annual Product Reviews
- Audit Reports (International and National Audits)
- Calibration Records of instruments
- CAPA (Corrective and Preventive Actions)
- Change Controls/ Deviations Records
- Cleaning and sanitization records
- Disposal Procedures and their Handling
- Product Dossiers
- Investigation Reports for Product Complaints if any
- Non-Conformance Reports (NCRs)
- Qualification documents like DQ, IQ, OQ and PQ of Equipment
- Vendor Approval records
- Sampling Procedures

- Training Evaluation Records
- Reports of water samples, air samples and other routine microbiological testing.

5.0 PRODUCTION

5.1 Brief description of production operations using wherever possible flowsheets and charts specifying important parameter.

The IV manufacturing plant uses a Form-Fill-Seal (FFS) technology, polypropylene (PP) film is used as the primary packaging material and is processed entirely in-line. Pharmaceutical-grade PP film is received as rolls, inspected, sampled, and released by Quality Control before use. During production, the film is continuously fed into the FFS machine, where it is heated and thermoformed into IV containers under controlled environmental conditions.

Once the container cavities are formed, the pre-prepared and filtered IV solution is accurately filled into the containers. The formed PP ply is then ported and heat-sealed to close the containers, producing a hermetically sealed unit without manual handling. Because the forming, filling, and sealing steps occur in a single enclosed system, the risk of contamination is minimized.

After sealing, the filled PP containers are subjected to terminal moist heat sterilisation in validated autoclave cycles to achieve the required sterility assurance level. Following sterilisation and cooling, the units are visually inspected for seal integrity, leaks, deformation, and particulate matter before labelling, secondary packaging, and final QA release. This approach ensures consistent product sterility and container integrity while relying on process control and terminal sterilisation rather than fully aseptic operations.

In the oral liquid manufacturing plant, production operations are organized to ensure consistent product quality, microbial control, and compliance with GMP, while recognizing that products are non-sterile and rely on formulation design and hygienic processing rather than sterilisation. Raw materials, including active pharmaceutical ingredients, excipients, flavours, and colours, are received, sampled, tested, and released by Quality Control before use. Approved materials are weighed and

dispensed in controlled dispensing areas according to the batch manufacturing record.

Solution or suspension preparation is carried out in stainless steel mixing vessels using purified water. Excipients are dissolved or dispersed in a defined sequence, followed by addition of the API. Mixing parameters such as temperature, agitation speed, and time are controlled to ensure homogeneity. In-process checks, including pH, viscosity, specific gravity, and appearance, are performed during compounding.

The bulk product is transferred through dedicated, cleaned, and sanitized pipelines to the filling area. Where required, the product may be passed through polishing filters to remove extraneous matter, with emphasis on bioburden control rather than sterilisation. Filling is performed on liquid filling lines into cleaned and dried bottles, typically under controlled but non-aseptic environmental conditions. Immediately after filling, containers are closed using appropriate caps. Filled units undergo in-process checks for fill volume, closure integrity, and container appearance. The products are then labelled, batch-coded, and packed into secondary packaging.

Finished products are quarantined pending completion of Quality Control testing and review of batch documentation. Final release is performed by Quality Assurance only after all specifications are met, ensuring the oral liquid products are safe, stable, and compliant for patient use.

(see annexure 9 for production flowchart)

5.2 Production details and capacity.

The primary packing material used for the intravenous infusion is the polypropylene pouch, found to be safer than previously used material (see report in annexure 9). With Oral liquids, the PET (polyethylene terephthalate) bottles are used to hold these medications, common in the industry.

The manufacturing capacity of the large volume parenteral is 15,000,000 pouches per annum and the oral liquid plant has a capacity 1,600,000 litres per annum.

These facilities are briefly described as follows:

All manufacturing and packaging operations are carried out under controlled environmental conditions in accordance with approved Standard Operating Procedures. Each stage of manufacturing and packaging is fully documented, and details of operations are recorded in controlled Batch Manufacturing Records.

All manufacturing activities and in-process controls are performed under the direct supervision of competent and qualified technical personnel to ensure compliance with GMP requirements. Dedicated and clearly demarcated areas are provided for the storage of raw materials, packaging materials, in-process materials, and finished goods, in line with their respective status. All mixing vessels, storage containers, and major production equipment used for batch manufacture are uniquely identified and appropriately labelled to ensure traceability and prevent mix-ups.

Prior to manufacturing, raw materials are sampled and tested by the Quality Control department. Only materials released by QC are issued for production. Approved in-process bulk products are used for filling and packing operations. Samples of finished products are tested as per approved specifications, and batch records along with analytical results are reviewed by the Head of Quality Control. Final release of each product for sale and distribution is authorized by the Quality Assurance Manager.

5.3 Arrangements for the handling of starting materials, packaging materials, bulk and finished products, including sampling, quarantine, release and storage:

The handling of materials and products is systematically planned, controlled, and documented to ensure effective testing of incoming materials, in-process samples, and finished products. All analytical methods used for testing are validated, where applicable, in accordance with regulatory requirements. Each container of active pharmaceutical ingredient is sampled and tested for identity. The Quality Control department is responsible for the testing and release of starting materials, packaging materials, in-process materials, and finished products. Standard Operating Procedures are established for material handling at every stage, including receipt, quarantine, sampling, testing, storage, and release. Approved materials are stored in segregated

areas with clear, color-coded labelling to reflect their status and to prevent mix-ups or cross-contamination.

Material identification and analysis are performed to confirm compliance with approved specifications. All in-process and final product tests are predefined and carried out as per approved test plans. Documented test methods and inspection procedures are maintained to verify each critical quality attribute.

The flow diagram covering receipt, handling, testing, and storage of raw materials and packaging materials is provided in Annexure 9.

Additional considerations for handling inprocess and finished products:

Any deviation from the approved manufacturing process is treated as a process deviation and is documented, evaluated, and approved by the Quality Assurance department. Corrective and preventive actions arising from such deviations are recorded as applicable. Standard Operating Procedures describe the sampling of in-process and finished products. Sampling activities are performed by In-process Quality Assurance personnel. During manufacturing, in-process materials and finished products are stored in designated quarantine areas identified for in-process and finished goods.

To prevent contamination or mix-up risks arising from residues of previous batches, including products, materials, or labels, line clearance is performed by Quality Assurance personnel prior to the start of any manufacturing or packing operation. This ensures that the work area, equipment, and production lines are clean and free from previous batch remnants.

Considerations for quarantine, release and storage of finished products.

Finished products are quarantined by the Production department until clearance is granted by the Quality Control department. Final inspection and testing are conducted to confirm compliance with approved specifications and procedures. The Quality

Assurance department reviews the completed batch manufacturing and packaging records, and the Head of Quality Assurance authorizes product release only after ensuring that all required test results are satisfactory and the batch has been approved by Quality Control. Until release, finished products are stored under controlled conditions to maintain their physical and chemical quality attributes.

5.4 Arrangements for handling of rejected materials and products

Rejected materials are kept in an isolated area dully labelled 'Rejected' and will be stored in rejected area assigned for the same.

The rejection reports are issued to the Stores by Quality Control, and these materials are then shifted from quarantine to rejected area. Rejected printed packaging materials are destroyed and not returned to the supplier. All destruction activity is done in the presence of Quality Assurance and Quality Control personnel.

5.5 Brief description of general policy for process validation

Process validation is taken to demonstrate, and to provide documented evidence that a process shall, with a high degree of assurance, consistently achieve the desired and intended results. Before process validation the term equipment qualification is an essential as a pre-validation phase. The validation policy of the company is to keep its manufacturing facility with utilities, critical equipment, manufacturing processes, cleaning and disinfection procedures and analytical methods in a validated state so as to produce drug products of consistent quality, as per cGMP norms, with high degree of confidence with established documented evidence.

Process validation of critical operations like manufacturing area, filling and sealing area is carried out with respect to VMP.

Following types of validations are carried out for the processes:

Prospective Validation:

- During product development, the production process is broken down into individual steps.
- Each step is evaluated to determine the critical factors/parameters that may affect the quality of the finished product.

- A series of experiments are planned and documented fully. A protocol based prospective validation includes the following elements:
  - Description of the process
  - Details of the equipment / facilities to be used
  - Identification of Critical Steps
  - Description of the validation plan- Sampling program/ plan, test plan
  - The acceptance limits and the time schedule.
  - Details of methods for recording and evaluating results, including statistical analysis.
  - Conclusions and recommendations.

Retrospective Validation

Annual Product Review fulfills the requirement that applies to existing processes and equipment and is based solely on historical information. Unless sufficiently detailed past processing and control records are available. For example, it is necessary to establish that the process has not been modified and that the equipment was still operating under the same conditions of construction and the performance as documented in the historical records.

Concurrent Validation

Concurrent validation is also carried out on the processes which are better understood, to establish documented evidence that a facility and process do what is supposed to do, based on the generated evidence during actual implementation of the process. This is the primary kind of validation used in the Biomedical Limited.

6.0 QUALITY CONTROL

6.1 Description of the Quality Control system and of the activities of the Quality Control department. Procedures for the release of finished products.

The Quality Control department is responsible for testing and evaluating the quality of incoming materials, in-process materials, and finished products in accordance with approved specifications. All testing activities are performed using validated or approved analytical methods.

To support these activities, the Quality Control department is equipped with appropriate and sophisticated analytical instruments and facilities. The department

comprises a chemical laboratory, a pyrogen testing laboratory, and a microbiological laboratory. Raw materials, packaging materials, in-process samples, and finished products are tested in accordance with approved pharmacopeial or in-house specifications and test procedures. The Quality Control department also conducts testing and evaluation of various packaging components to ensure their suitability for use. Microbiological analysis of water systems, manufacturing environment, starting materials, in-process samples, and finished products is carried out by the microbiology section of the Quality Control department.

In addition, the Quality Control department is responsible for post-production stability studies of manufactured batches, which are conducted as per approved stability protocols in compliance with applicable ICH guidelines.

7.0 CONTRACT ACTIVITIES

7.1 Description of the way in which the GMP compliance of the contract acceptor is assessed

At present, no loan license manufacturing activities are being carried out at the facility. If such activities are undertaken in the future, they shall be performed at the contract acceptor's site only after execution of a formal written agreement between the contract giver and the contract acceptor, clearly defining the responsibilities and duties of each party in accordance with GMP requirements, NAFDAC regulations and WHO guidelines. Ultimately, the responsibility for product quality, safety, and regulatory compliance remains with Biomedical Limited.

Loan license manufacturing shall be conducted in a manner that ensures consistent product quality, safety, and compliance with regulatory standards. The GMP compliance status of the contract acceptor shall be assessed through qualification and routine audits conducted in accordance with approved Standard Operating Procedures for monitoring of contract manufacturers. Audit findings, corrective actions, and follow-up activities shall be documented and reviewed to ensure ongoing compliance.

7.2 Types of Contracted Activities

Contracted activities may include, but are not limited to:

- Contract testing of raw materials, packaging materials, in-process samples, or finished products.
- Specialized analytical testing not available on-site, such as advanced microbiological or chemical testing.
- Calibration and qualification of critical equipment and instruments.
- Maintenance and validation of utilities such as water systems, HVAC systems, and cleanroom monitoring and qualification.
- Pest control services within production, warehouse, production and laboratory areas.
- Contract manufacturing or packaging, where applicable.

7.3 Selection and Qualifications of Contractors

All contractors are selected based on their technical competence, regulatory compliance history, and ability to meet GMP requirements. Contractor qualification includes:

- Evaluation of company profile, licenses, and certifications
- GMP compliance assessment through questionnaires and, where applicable, on-site audits
- Review of personnel qualifications, facilities, equipment, and quality systems
- Approval by the Quality Assurance department prior to engagement

Records of contractor evaluation and approval are maintained.

7.4 Technical and Quality Agreements

All contract activities are governed by written Technical and/or Quality Agreements that clearly define the roles, responsibilities, and GMP obligations of both parties. These agreements include:

- Scope of contracted activities
- Responsibilities for sampling, testing, documentation, and data integrity
- Change control and deviation handling responsibilities
- Communication pathways and reporting timelines
- Data ownership, confidentiality, and retention requirements
- Audit rights and regulatory inspection readiness

No contract activity is initiated without an approved agreement in place.

7.5 Oversight and Performance Monitoring

Biomedical Limited maintains ongoing oversight of all contract activities through:

- Periodic performance reviews
- Review and approval of test results, certificates of analysis, and service reports
- Monitoring of deviations, out-of-specification results, and CAPA implementation
- Scheduled requalification and re-auditing of contractors, based on risk assessment

The oversight team, usually consisting of the Quality Assurance, Engineering and Production Managers, reviews all outputs from contract activities before they are accepted for GMP use.

7.6 Change Management and Deviations

Any changes proposed by the contractor that may impact product quality, regulatory status, or GMP compliance are subject to Biomedical Limited's change control system. Deviations arising from contract activities are documented, investigated, and closed with appropriate CAPA in coordination with the contractor.

7.7 Complaints, Recalls and Regulatory Interactions.

Contractors are required to immediately notify Biomedical Limited of any quality issues, complaints, or regulatory actions that may affect contracted activities. Contractors shall support investigations, recalls, and regulatory inspections as required. Biomedical Limited retains authority for communication with regulatory agencies.

7.8 Records and Documentation

All records generated from contract activities are considered GMP documents and are retained according to Biomedical Limited's document retention policy. Data integrity principles, including ALCOA+, are applied to all contracted activities.

8.0 DISTRIBUTION, COMPLAINTS AND PRODUCT RECALL

8.1 Arrangement for distribution:

Finished products, after approval for sale and distribution, are transferred from the quarantine store to the approved finished goods store for onward distribution. Finished goods are properly segregated, clearly identified, and stored on pallets or racks with appropriate labelling to prevent mix-ups and ensure traceability.

Before dispatch, store personnel verify that the finished products have been released by the Head of Quality Assurance. Packed goods are dispatched to distributors or customers through approved contract carriers. All dispatches are carried out strictly on a FEFO basis (First Expired First Out) and on a batch-wise basis only.

Comprehensive distribution records are maintained to ensure complete traceability of each batch of every product throughout the distribution chain.

8.2 Arrangement for the handling of complaints and product recalls.

Quality Assurance is responsible for the receipt, handling, and investigation of all product complaints. An approved Standard Operating Procedure is in place for the management of product complaints and product recalls. Each complaint is investigated in accordance with written procedures, and appropriate corrective and preventive actions are identified, implemented, and documented. Records of market complaints are retained for a period of three years after the expiry of the respective products.

If a product recall is required, the Head of Quality Assurance acts as the Product Recall Coordinator and is responsible for coordinating and overseeing all recall activities. Product recalls are conducted in accordance with approved written procedures that clearly define the sequence of actions to be followed.

Approved procedures are available for retrieval of distribution and sales data, notification of customers, receipt and segregation of recalled products, inspection and investigation of the recalled material, and implementation of corrective actions. Depending on the nature and classification of the recall, wholesalers, distributors, retailers, and medical practitioners are involved as appropriate. The Recall Coordinator ensures that the competent regulatory authority is informed of the recall in a timely manner.

9.0 SELF INSPECTION AND CONTINUAL IMPROVEMENT

Self-inspections, also referred to as internal audits, are planned and conducted at defined intervals to monitor the implementation and continued compliance with current Good Manufacturing Practices. Self-inspections are performed in accordance with the approved Standard Operating Procedure for internal audits and self-inspection.

Internal audits are carried out either by a designated internal audit team or by authorized Quality Assurance personnel. Members of the internal audit team are selected from various departments, excluding the department being audited, to ensure objectivity and independence. All inspections are documented and compiled into a formal audit report.

The audit report includes details of observed deficiencies, proposed corrective actions, the personnel responsible for implementation, and defined compliance timelines. The report is communicated to the heads of the concerned departments. Follow-up activities are conducted against the agreed compliance dates to verify that corrective actions have been effectively implemented and closed.

(see annexure 10 for list of GMP inspections)

----------END OF SMF DOCUMENT----------